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Refracturing Success Study in Medium Permeability Conventional Reservoirs

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Abstract

Hydraulic fracturing operation is often the most efficient method to develop low-to-mid permeability reservoirs. However, not every reservoir and not every fracturing operation have the same longevity of the positive impact. Re-fracturing operations have proven to be a success on many occasions, however industry still lacks a consistent guideline of when, how and where to perform re-fracturing treatments. This study provides in-depth research of refracturing methods for mid-permeability conventional oil reservoirs.

Study represents an integrated approach covering all aspects of brown field development via re-stimulation that included candidate selection; verification of remaining oil-in-place; estimating the current skin-factor of the well through sophisticated workflow of production history match and initial fracturing geometry match through pressure behavior; assessing the risk of water breakthrough from underlying layers or waterfront from offset injector wells. The methodology to optimize refracturing operation was formulated through design-execution-evaluation cycle with detailed workflows across each phase. Mathematical normalization methods were used to compare different wells under common grounds for efficiency identification.

Statistical analyses were based on refracturing operations in more than 30 wells from 9 oil fields and multiple reservoirs. The study revealed that even under-designed initial fracturing treatments could result in a skin-factor of -3.5 to -4.0 and throughout the period of 2-10 years' time more than half of the wells could maintain the negative skin factor of an average of -1.8 at the time of refracturing operation. However, the refracturing operation could still gain significant benefit and improve the final skin factor up to -5.1 when a careful candidate recognition process was followed. Four case studies with different scenarios are given to showcase the process to decide of refracturing and succeed through the design-execution-evaluation cycle with the detailed workflow. As shown in the paper, the integrated approach helped engineering team to enhance the incremental oil production from refracturing operations by 43% when compared to conventional refracturing approach; and respectively increased economics success ratio from 34% to 79% based on normalized cash flow after 270 days of post-refracturing cumulative oil production.

Study represents one of the few documented cases of massive refracturing operations in medium permeability reservoirs. Mathematical methods used to visualize and normalize the production history for the representative comparison and workflow optimization. Comprehensive decision trees and refracturing

flow charts will be valuable for a wide audience of engineers involved in mid-permeability formations development through hydraulic fracturing methods. This discipline is expected to gain momentum in the Middle East soon.

Introduction

Refracturing operations in mature oil fields have emerged as a crucial technique to enhance hydrocarbon recovery and extend the productive life of wells. These operations are particularly effective in fields with medium permeability, where initial fracturing treatments may not have fully optimized reservoir stimulation. A significant amount of remaining oil in place (ROIP) often exists in these reservoirs, making them prime candidates for refracturing. In medium permeability sandstone formations, refracturing can significantly improve performance by reactivating existing fractures and creating new pathways for oil flow. Dual productivity index (PI) analyses are essential in evaluating the effectiveness of refracturing treatments. These analyses help in understanding the changes in well productivity before and after refracturing, providing insights into the success of the operation as shown by [Ren et al. 2024](#).

Candidate selection for refracturing is a critical step that involves evaluating wells based on various criteria, including initial production decline, reservoir depletion, and the quality of the initial completion. Wells with ineffective initial completions or those located in underdepleted reservoirs often show the most promise for successful refracturing as described by [Roussel et al. 2013](#). [Yan et al. 2024](#) described an example of implementing set of advanced diagnostic techniques, such as distributed temperature sensing (DTS) and real-time reservoir modeling, that play a vital role in identifying optimal zones for refracturing.

Recent studies have highlighted the importance of integrated workflows that combine geological and engineering data to optimize refracturing designs. For instance, a study by [Ren et al. \(2024\)](#) emphasizes the need for accurate modeling of fracture propagation and dynamic stress changes during production ([Gharawi et al. 2022](#)). Overall, refracturing operations in mature oil fields with medium permeability offer a strategic approach to enhance hydrocarbon recovery and extend well life.

The paper will describe the case study from Western Siberia's clastic reservoirs. The oil province is mature and many fields require re-stimulation approach to stay economical. A detailed review of the geology, fracturing operations and some novel stimulation techniques in the particular area were earlier provided by [Yudin et al. 2013](#); [Doctor, et al. 2014](#); and [Valiullin et al. 2015](#).

By leveraging advanced techniques and integrated workflows, operators can optimize refracturing designs and achieve sustainable production improvements. Following chapters will demonstrate a successful case study performed with refracturing campaign in Western Siberia onto mature fields. The paper will not describe details on well completion techniques, as it mainly focuses on vertical wells cases. However, when it comes to horizontal multistage operations in mature fields of Western Siberia, extensive research and solutions were offered by [Yudin et al. 2014](#), [Butula et al. 2015](#) and [Serdyuk et al. 2016](#).

Geology Overview

Most of the studied oil fields have several producing horizons - Jurassic formation, BV 18-22 (Achimov formation) and upper BV and AV formations of Cretaceous age. All formations are sandstones, producing oil. Most fracturing treatments were performed on Jurassic formation, which is a deepest one in the area.

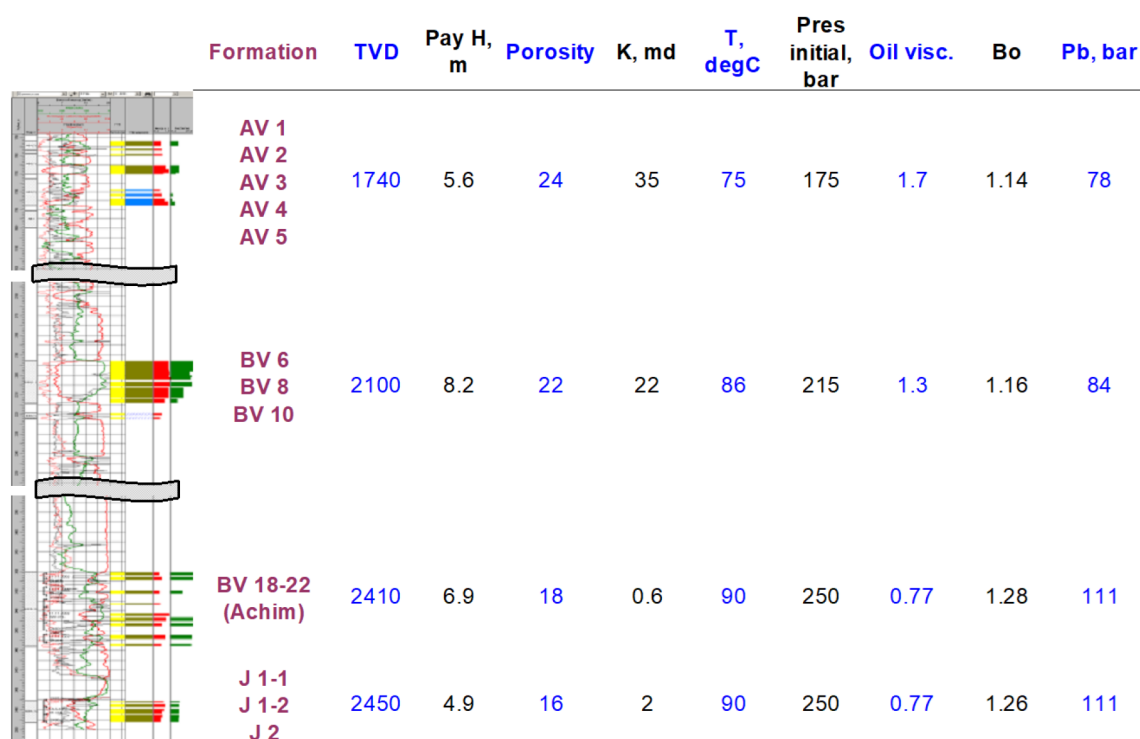


Figure 1—Stratigraphy and formation properties of the representative well in Field V.

Tectonically wise the area locates on the northern slope of Nizhnevartovsk dome and in the eastern part of dome-shaped high, which complexes the dome. The dome has got a slightly stretched in meridian direction form with dissected contour. The dome is 160 km wide in its northern part and it is essentially receding to the south. The dome is 250 km long. Agan high is a subsidiary structure; high amplitude composes 135 meters; flanks slope angle is up to 70 meters. In Jurassic reservoir for example, a high is represented by four dome-shaped highs 8,8x7 km, 4,4x1,6 km, 2,8x1,6 km in size. Structural plans of horizons BV18-21(Achimov) repeat Jurassic structural plan and mainly match with seismic horizon. Structure's amplitude is 90 m and it is 22x25 km in size. Structural plan is vertically flattening to the top. Amplitude of Cretaceous formations is reduced to 35 m.

Jurassic, Cretaceous, Paleogene and Quaternary depositions occur in cross-section of Mezzo-Cenozoic terrigenous depositions of the platform mantle. 13 net horizons are identified in Field cross-section of Low-Cretaceous depositions (from the top to the bottom): AV13 of Low-Alymsk formation, Aptian Stage; AV3, AV4, AV5, BV1, BV2, BV3, BV6 of Vartovsk formation; BV8, BV9, BV91, BV18-19, BV20-21 of Megion formation Barremian, Hauterivian, Valanginian and Berriasian Stages. JV1 net horizon occurs in Upper-Jurassic depositions. Oil-bearing layers AV4-5 and BV1-3 were discovered in the drilling process.

First 10 years of the hydraulic fracturing operations in the area utilized a very conservative design – from 5 to 10 metric ton of proppant with high polymer concentration damaging the formation. Most of the wells were treated with only 6-10 tons of proppant. That time service companies did not implement calibration stage, all fracturing operations consisted of big pad volume (around 30 m³) and conservative proppant placement (upto 850 KgPA or 8PPA), low amount of breakers was used. All that created an opportunity for refracturing, with the main goal to increasing production by performing better quality jobs.

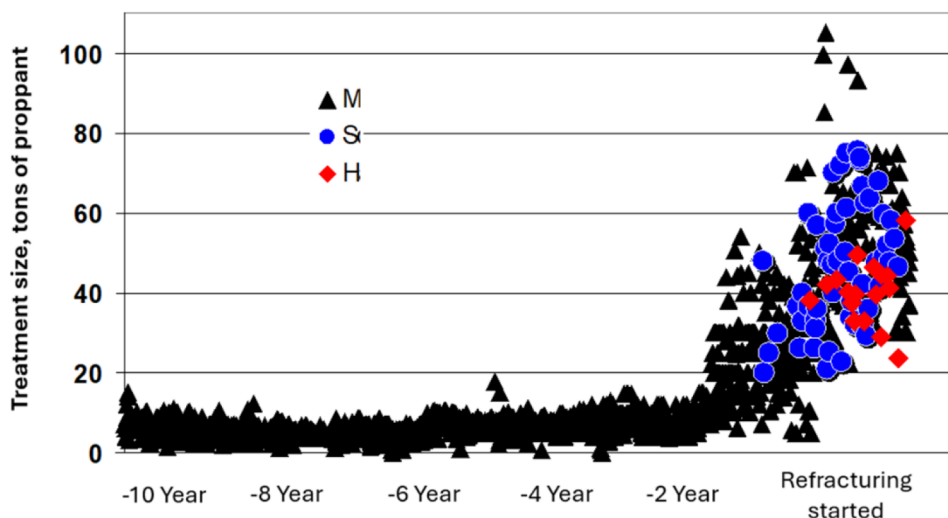


Figure 2—Historical overview of performed fracturing operations – basic designs during first 10 years and enhanced design quality currently. Quality was increased due to competition induced by the operator (see different service companies in different colors). A refracturing campaign has also started as part of the enhanced stimulation program.

Fig. 3 represents a historical overview of the fracturing activity in the area. As mentioned, during the first ten years of fracturing activity there were about 100-150 fracturing operations performed each year in the old producing wells. Only recently operator started to implement hydraulic fracturing in newly drilled wells in tight areas since production without stimulation did not prove economical. Last three years – as it was also noted from the previous plot a massive campaign started utilizing resources of 3 service companies and including re-fracturing operations on those wells previously fractured within last 10 years.

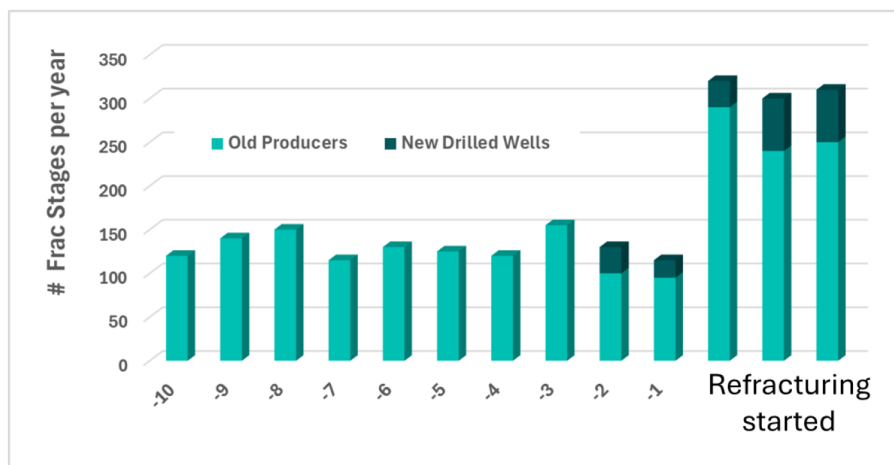


Figure 3—Stage count of fracturing activity – prior the enhanced stimulation campaign and during it.

Fig. 4 gives a distribution between various oil fields in the area – in terms of total number of producing wells and also in terms of wells fractured per oil field. Overall, more than 20 oil fields were studied in the project but only 8 of them were selected for this paper as being the most representative ones and having high percentage of fracturing activity.

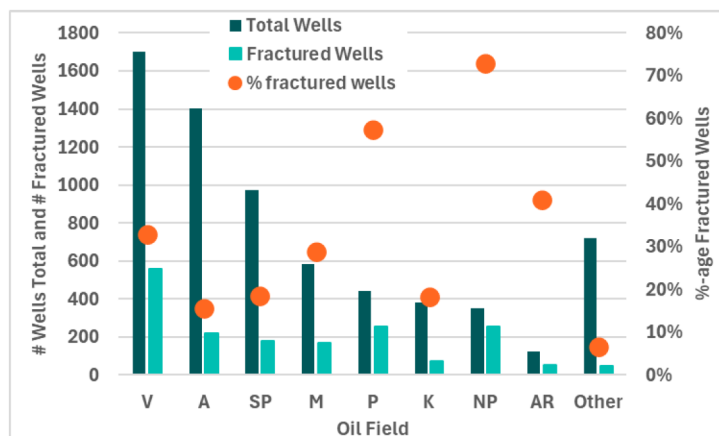


Figure 5—Stage count of fracturing activity – prior the enhanced stimulation campaign and during it.

Refracturing Candidate Selection

To calculate the remaining producible oil, one should start with the Original Oil in Place (OOIP), which is determined using the volumetric method:

$$\text{OOIP} = 7758 \times A \times h \times \phi \times (1 - S_w) / B_o$$

where A is the drainage area, h is the net pay thickness, ϕ is the porosity, S_w is the water saturation, and B_o is the formation volume factor.

Next, we account for the cumulative oil production to date (N_p) and the recovery factor (RF):

$$\text{Recoverable Oil} = \text{OOIP} \times RF$$

The remaining producible oil is then:

$$\text{Oil remaining to recover} = \text{Recoverable Oil} - N_p$$

In summary, the remaining producible oil is calculated by determining the OOIP, applying the recovery factor to estimate recoverable oil, and subtracting the cumulative oil production to date. This method provides a straightforward approach to estimate remaining reserves based on key reservoir parameters (see Fig. 6 to visualize excel calculator used to make such calculations). This was a very critical step in candidate selection prior to proceed with other criteria: we set the threshold for the refracturing candidates to have at least 20% of remaining producible oil.

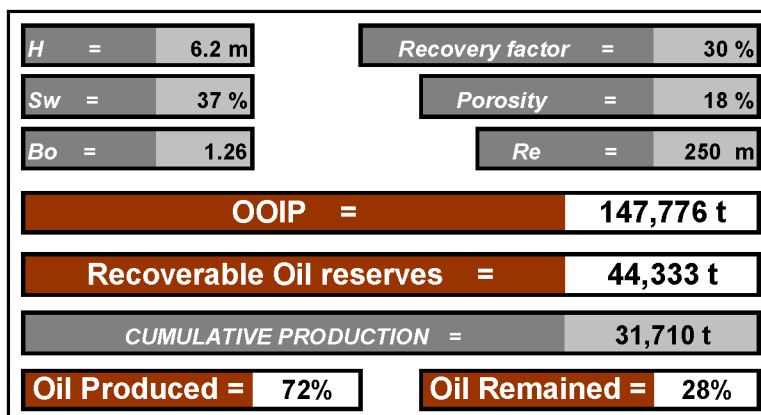


Figure 6—Remaining recoverable oil calculator based on reservoir parameters and cumulative produced oil inputs.

The main criterion for selecting candidate wells is the increase in oil production after treatment. Proposals showing less than 30 tons of oil per day (TOPD) of increased production were not considered as economically viable. Wells producing more than 20 TOPD before the refracturing operation were not considered as a candidate for treatment due to past negative experiences and risks to lose a good well (due to water breakthrough etc). While good wells are suitable for certain treatments, the focus was on finding candidates among the wells with lower current production but with high potential. This approach limits the selection and complicates the process, however it provide most reliable results for the well stimulation program. Most local reservoirs are depleted, and there is an intensive water re-injection system organized to maintain formation pressure at the economical level. However, efficiency of this formation maintenance system is not uniform – so it is of the ultimate importance to select the area of the refracturing candidate in respect to the current formation pressure but at the same time ensuring the new fracture will not break into water flood.

Accurate estimates of liquid production post-treatment are needed to use the correct artificial lift equipment. Three service companies are competing, and wells are evaluated by all three. The company predicting the highest increase in oil production gets the well. During monthly meetings, each service company defends their results. Apart from operator-suggested candidates, companies can propose their own. The service companies have access to databases, software, and networks, including production data, exploration maps, and logging information. Regular communication with representatives is common practice. This set up provides very strong teamwork and engineering excellence allowed technical experts from different disciplines work on the common goal of maximizing production potential of the field.

The candidate selection process during operations is divided into two parts, evaluated by different groups with varied expertise. One group focuses on reservoir engineering and production prediction, while the other group handles well production history, logs, offset wells production, well deviation, and cement quality. Software tools are used to simulate and optimize treatment geometry and predict production. Both groups agree that certain tools are valuable for identifying candidates, estimating fracture geometry, and predicting productivity. For wells not previously treated, identifying potential issues is complicated due to data uncertainty. Accurate permeability values are crucial for production prediction.

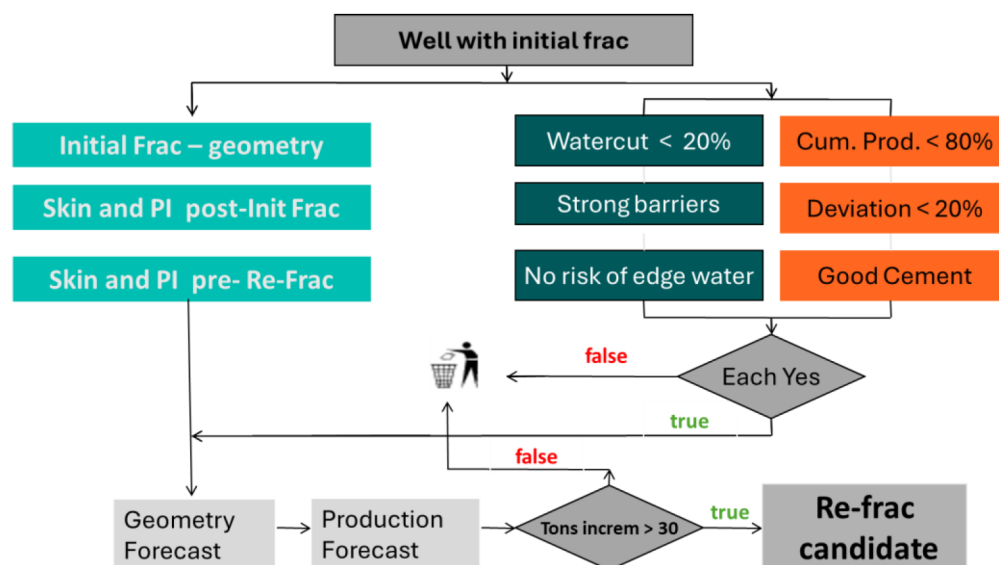


Figure 7—Candidate selection workflow.

The Design-Execute-Evaluate (DEE) engineering cycle for the refracturing operations was different from the initial fracturing cycle. Below is a summarized description of it.

Design Stage: The primary objective of re-fracturing is to achieve optimal fracture geometries. For formations with low permeability, the focus is on creating longer fractures, while for higher permeability formations, wider fractures are desired. Fracture reorientation is generally not a significant concern, except for ensuring proppant admittance in deviated wells. Fracture geometry simulations are conducted using historical data from previous treatment and give an advantage in predicting accuracy if current reservoir pressure and skin factor were possible to extract from production history match. Local guidelines suggest using 4-5 tons of proppant per meter of production interval, with a maximum concentration of 1100 kg/m³, adjusted for deviated wells to minimize the risk of wellbore screen out. Reperforation is a standard procedure, using specific perforation guns and phasing to ensure effective slurry channels.

Execution Stage: Pumping schedules include injection and calibration stages, with an average fluid efficiency of 50-60%. To avoid excessive polymer injection and formation damage, a maximum of 10-15 m³ of pad is used. Additional breakers are pumped to address existing damage from previous treatments and to clean out new fractures. High fluid efficiency treatments often use techniques to restrict fracture height growth and increase propped width. For example, a specific technology (so-called tag-frac) is used to add a stage that causes early bridging in the shale, effectively containing the fracture within barriers and increasing its propped width.

Operational Considerations: Jobs are performed by a fleet consisting of multiple units of equipment and crew members. The geographical spread of oil fields and the presence of rivers pose significant challenges to mobility and logistics. Emphasis is placed on service quality, field-office communication, and regular training. One operational mistake can jeopardize the whole process and fail the well to produce as per expectation – so highly motivated and experienced personnel is a must. Equipment also needs to be properly maintained and function tests to be done prior to every stage.

Evaluation Stage: Pressure matching is performed to determine refracturing geometry. Leak-off and stress profiles may vary between old and new fracture areas, affecting zone characteristics. For example, during initial treatments, proppant may settle at the bottom of the fracture, leading to partial zone coverage. During re-fracturing, the fracture may grow within the top part of the formation before covering the entire interval. Production forecasts are adjusted based on the obtained fracture geometry, though these adjustments are typically minor. Post-job reports are created, and incremental oil production is monitored and reported at monthly meetings. Post-treatment production monitoring provides valuable information for investigating fracture performance. This stage also allows closing the DEE cycle loop and learning from it for the next wells.

The following section will describe the process of DEE implementation to 2 wells (M-1 and V-1) in different fields to describe challenges and solutions of refracturing operations in more details.

Unsuccessful Refracturing Case and Lessons Learned. Well M-1

Well M-1 was one of the initial wells in the campaign and helped to learn several lessons building the successful workflows – therefore it is given here as a negative example. The well was originally fractured right after drilling before put into production. However, oil rate dropped to uneconomical values within the first 10 months of production, as it can be seen from Fig. 8. To restore the potential, a refracturing (RF) operation was pumped using larger volumes. However, despite of fluid production increase, oil incremental gain was only 9 tons a day due to increased water cut and the treatment was not profitable.

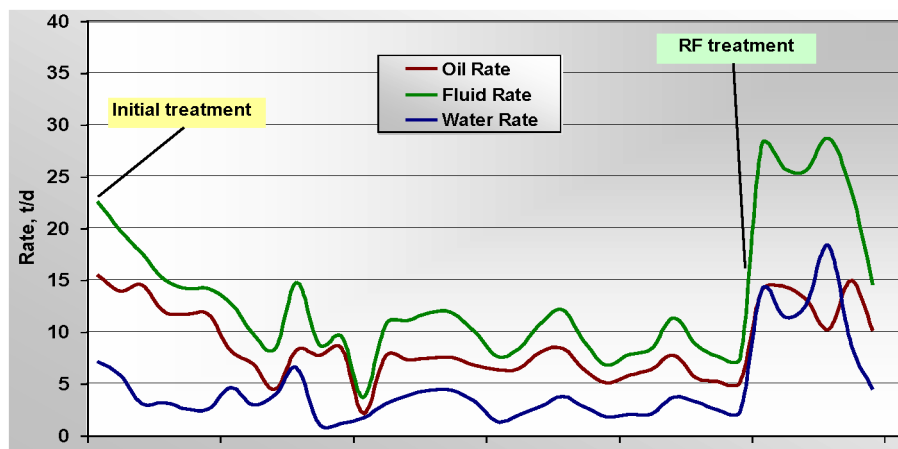


Figure 8—Well M-1 production history – after initial fracturing and after refracturing (RF) treatment.

As it can be seen from Fig. 9, Jurassic formation in the area has oil saturation on top and water saturation at the bottom, which makes fracturing operation risky. Therefore the initial treatment was designed with lower volumes to minimize risks of vertical growth. This limitation was not carefully considered for the selection of the best refracturing candidates. The presence of water below may either limit the treatment size and therefore restrict the fracture length or it will mean the risk of breakthrough into the water layer leading to excessively high water cut.

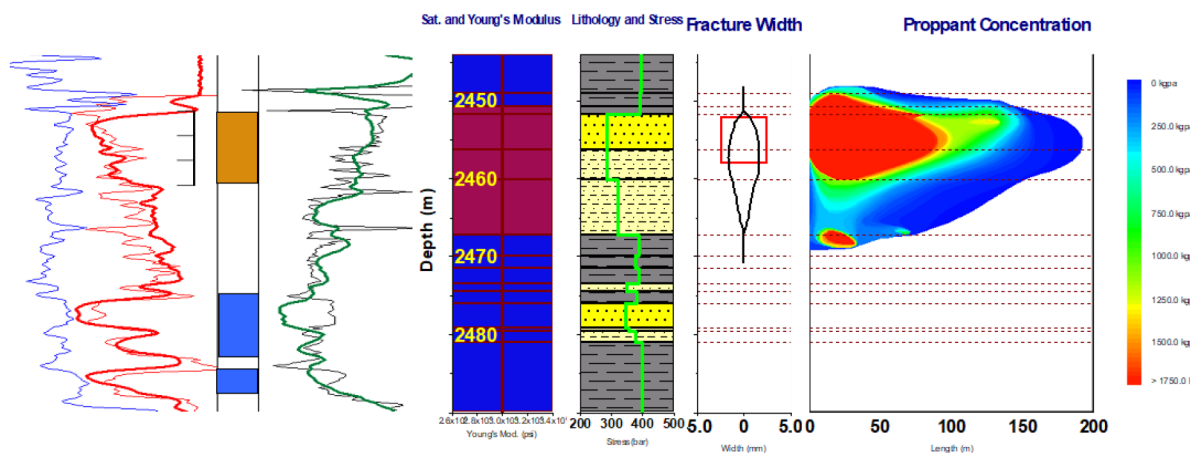


Figure 9—Geological cross-section and initial fracture geometry avoiding water zone breakthrough.

Next step of investigation was to carefully evaluate current state of productivity index (PI) and reservoir pressure (Pr) – as shown in Fig. 10, a) and b) respectively. A representative set of points normally selected for this exercise within 2-3 last months of production prior to refracturing operation. While PI was lower than the offset wells, the reservoir pressure extrapolated from representative points of production was somewhat depleted suggesting the existing skin factor remaining from original fracture was around -2.4. The assumption to design a refracturing treatment originally was about zero skin. Despite refracturing treatment improved the skin factor down to -5.5 due to increased proppant mass used when compared to initial fracture, the oil incremental rate was not sufficient due to increased in water cut. Presence of water injector well in the direction of fracture propagation could also cause the water cut to increase considering a possibility of the frac to break into injected water zone.

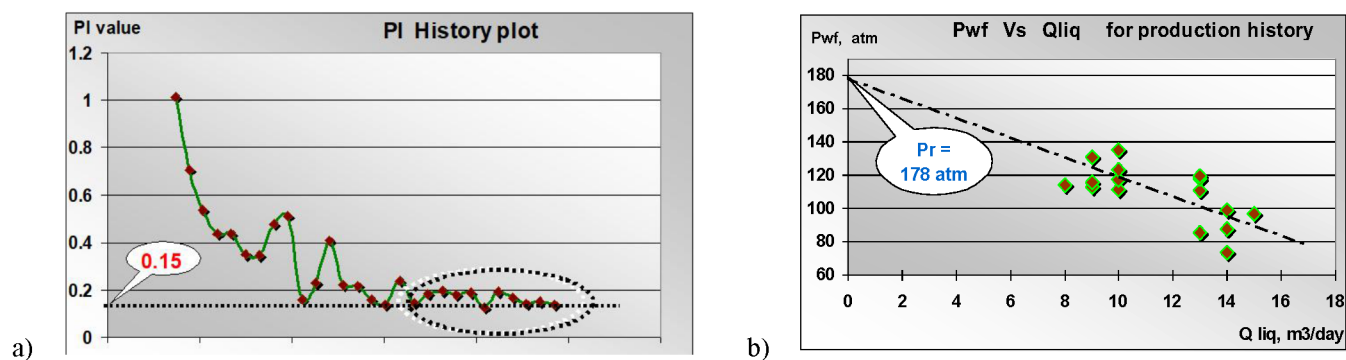


Figure 10—Calculating current productivity index (PI) and reservoir pressure (Pr) based on production history match in well M-1.

The hydraulic fracturing operation itself was performed with high quality and in aggressive manner – having minimum formation damage (small pad stage and aggressive proppant ramp) and significant breakers concentrations, as can be seen from pressure, rate and concentration (PRC) plot in Fig. 11.

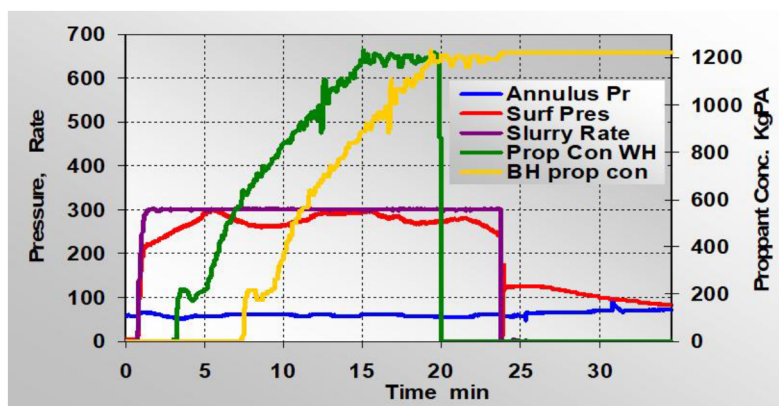


Figure 11—Refracturing operation PRC plot in well M-1.

This case study confirms that even with the high-quality design and execution – hydraulic fracturing treatment does not guarantee success of the production if the risks are not considered and geological uncertainties are not properly addressed.

Successful Refracturing Case. Well V-1

Well V-1 was a typical well treated with refracturing during the enhanced stimulation campaign once all the proper workflows established – and given here as a representative case study. As can be seen from Fig. 12 – the original fracture quality was very conservative and effective fracture length was estimated to be only 30–40 meters. Original skin factor after the operation was evaluated around -2.4 based on production history match. While selecting the well for the refracturing operation, its current skin was close to zero (-0.4) making it a potential candidate provided there was no risks with vertical breakthrough into water, no injectors water nearby, decent cement quality, no screen out risks, sufficient remaining oil to produce and all the other factors described in the previous section were favoring this candidate for a refracturing operation.

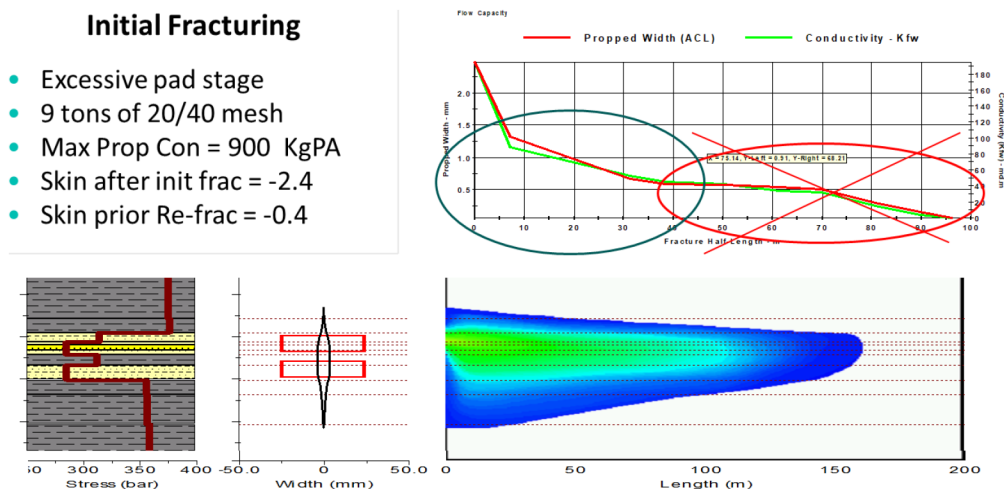


Figure 12—Summary of the initial fracturing in well V-1.

The refracturing treatment itself was executed successfully with high service quality and adherence to operational and safety standards. The treatment plot is given on Fig. 13. One can observe successful pressure test, injectivity test, calibration treatment followed by redesign and the main treatment performed with minimum polymers concentration and aggressive proppant ramp. So-called tag-frac method was used implementing proppant tag during the pad to restrict vertical fracture growth (bridging into shaly barriers) and therefore increased the width of the fracture – to improve conductivity and post-refracturing production. Increased amount of breakers were used to destroy previously induced polymer residues of the initial frac.

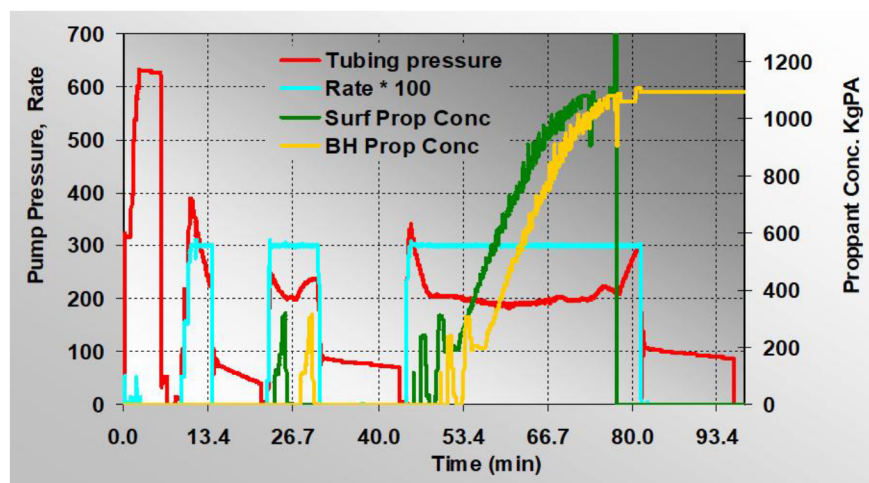


Figure 13—Summary of the initial fracturing in well V-1.

Fig. 14 provides a summary of the refracturing geometry and parameters – utilizing 5 times larger treatment when compared to initial operation – with bigger mesh size, providing more than 100 meters of effective fracture length and resulting skin factor of -5.4 (which was confirmed by post-refracturing production history match).

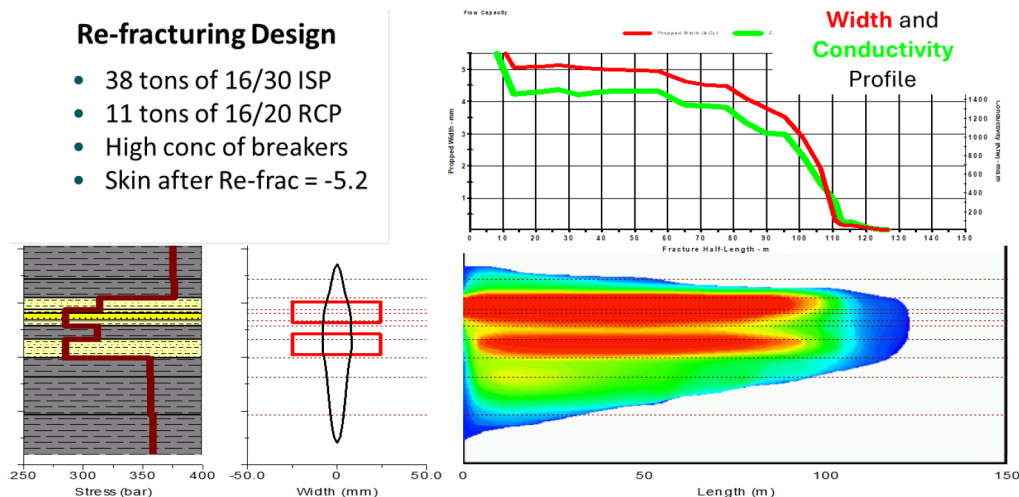


Figure 14—Summary of the re-fracturing in well V-1.

Fig. 15 shows the overall history of the well V-1. Without economic production after drilling, well was idle for several months. Then the initial frac (IF) was successful. An electric submersible pump of 20 m³/day capacity (ESP-20) was installed immediately after it. Then an ESP change for bigger capacity (ESP-50) was attempted showing success for almost a year period. However, the sudden drop in productivity made operator change the artificial lift to low-producing sucker-rod pump. After another year of slow production, the decision was to perform wellbore cleanout operations and install ESP-30 in place that was successful for another year of production. Nevertheless, during the refracturing candidates screening the well was showing potential based on currently high reservoir pressure skin being around zero. As it can be seen from the picture, a refracturing treatment with high volume of proppant really made a difference – improving skin factor down to -5.2 and allowing to keep 50 TOPD at a stable manner and leading to 38 TOPD incremental oil production. One note to be made here talking about conductivity increase in depleted formations (as is often a case for refracturing operations): proppant flowback control can be a significant challenge. ElSebae et al. 2020 provided an overview of modern solutions to address these challenges.

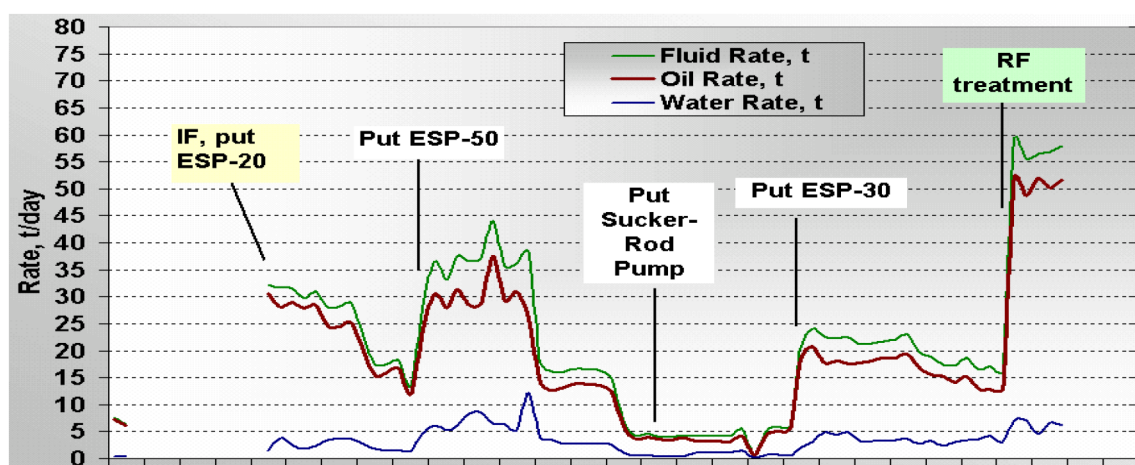


Figure 15—Production history of the well V-1 showing advantage of refracturing (RF) operation.

Refracturing Statistical Analyses

Overall statistical analyses of the refracturing campaign are given in Table 1. They are represented in terms of skin-factor. Model for every well was calibrated based on production history match in several representative points (original production when a reservoir pressure is known, re-fracturing operation where

post-frac skin can be estimated and when reservoir pressure is measured during injectivity test, and also re-fracturing operation. This approach allows deriving critical derivatives out of those three equations – kH value, skin original, skin pre- initial frac and skin pre-refracturing operation. This methodology called dual-PI analyses is a powerful tool and has multiple representative points such as in case of multiple interventions allows to reproduce productivity index with high accuracy (assuming kH value remains the same for entire well life in our case). These statistics can be used to estimate the potential of similar refracturing campaigns in other projects. Bottom row of the table can be given as an average statistical parameter: average skin post-drilling was 6.2; then further increased to 20.5 on average prior to initial fracture treatment. Initial fracture treatment was moderately efficient due to low treatment sizes and polymer damage, leading to skin of -3.5 on average. Most interesting point is a remaining skin-factor after years of production (it takes from 10 to 3 years between initial fracturing and refracturing for the most of wells studied in this research). And longevity of the fracture was mainly depended on the reservoir, not the treatment itself – thus, in Field A, K and many wells of field M we observe very high (bad) skin-factors of around 10 and higher prior to refracturing operation. In other fields – such as NP, P and SP the skin-factor did not deteriorate that substantial over the years. Those wells showing almost stable skin at significant negative values maintain for up to 10 years despite the low quality of the original frac. In all cases, refracturing operation has led to significant improvement of skin-factor when compared to initial fracture due to substantial increase in fracture quality. This allowed us to regain original oil production or quite often even increase it.

Table 1—Statistical data for the re-fracturing operations performed in 30+ wells providing skin-factors and comparing to initial fractures.

Well	Skin after drilling	skin pre IF	skin post IF	skin pre RF	skin post RF
A-1	0	20	-3.7	8.0	-5.3
A-2	2	24	-4.0	10.0	-5.2
A-3	8	40	-3.0	11.0	-4.7
A-4	25	25	-3.8	14.0	-4.5
A-5	35	35	-3.3	15.0	-5.1
K-1	16	15	-3.5	2.0	-5.1
M-1	N/A	N/A	-4.3	-2.4	-5.6
M-2	0.5	5.5	-3.8	-1.4	-5.4
M-3	2.4	19.5	-4.1	24.0	-5.5
M-4	N/A	N/A	-2.9	25.0	-5.0
M-5	24	24	-4.1	29.0	-5.3
M-6	8	100	-4.0	65.0	-5.1
NP-1	0	27	-4.8	-3.0	-5.2
NP-2	1.5	10	-4.4	-3.0	-5.1
NP-3	1	2	-3.4	-2.0	-5.3
NP-4	3.5	26	-3.4	-2.0	-4.3
NP-5	0	17	-3.6	-2.0	-5.0
NP-6	5	7	-3.6	-1.5	-5.7
NP-7	0	14.5	-2.7	-1.0	-5.5
NP-8	5	70	-3.7	-0.8	-5.0
NP-9	2	2	-3.5	0.6	-5.3
P-1	4	6	-3.5	-2.0	-5.0
P-2	N/A	N/A	-2.5	-2.0	-4.2
P-3	N/A	N/A	-3.0	-1.5	-4.6
P-4	0	10	-3.0	-1.5	-5.3
SP-1	5	4	-3.5	-2.2	-5.4
SP-2	3.3	6	-2.9	-2.2	-5.1
SP-3	6	6	-3.1	-1.8	-5.3
SP-4	2	6	-3.5	8.0	-5.2
V-1	10	10	-3.3	-0.4	-5.2
V-2	2.5	35	-2.5	0.5	-4.3
V-3	1	9	-3.6	20.0	-5.4
Average	6.2	20.6	-3.5	6.2	-5.1

Fig. 16 provides a comparison between initial fracturing efficiency and the one of refracturing campaign. The plot is based on average numbers of the productivity index (PI) from the analyses mentioned in

the previous chapter. Thus, one can observe a productivity gain of 3.7 folds as an average efficiency of refracturing operation, which is a definite advantage of hydraulic fracture treatment when performed properly. Still, one needs to ensure adequate reservoir pressure, sufficient recoverable reserves in place remained and risk-free operation in terms of water cut increase to ensure economical justification for the refracturing treatment.

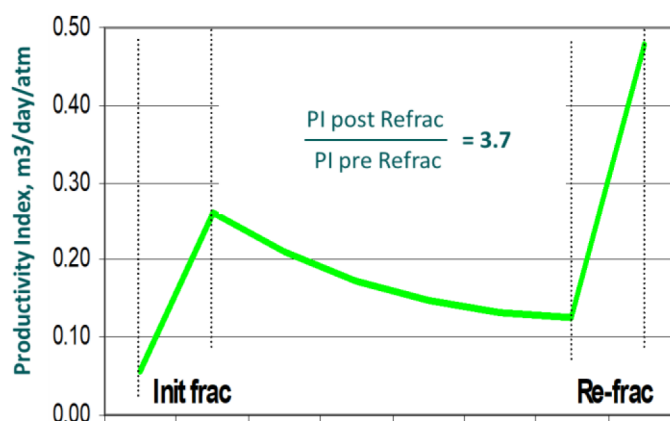


Figure 16—Comparing productivity index (PI) of initial fracture and refracturing operations.

Fig. 17 gives a normalized plot for the visualization purposes – all treated wells were put together on a single plot with averaged parameters, based on linear normalization of time scale for each of three periods – before initial fracture, in between the fracturing operations and post-refracturing. In such manner, it can be concluded from the plot that initial fracturing was successful, however it was not unlocking the reservoir up to its full potential – as it can be concluded from seeing the result of refracturing operations performed with increased proppant mass and optimized fluid design. On average, we can conclude an incremental oil production of more than 30 TOPD at the first month and about 10-12 TOPD per well once the production stabilized after 6 month. This incremental oil production can pay out the investment in refracturing campaign within 6-8 months based on economic analyses which is a viable investment considering a long-lasting effect after the fracturing treatment (although subject to case-by-case verification for each field and formation).

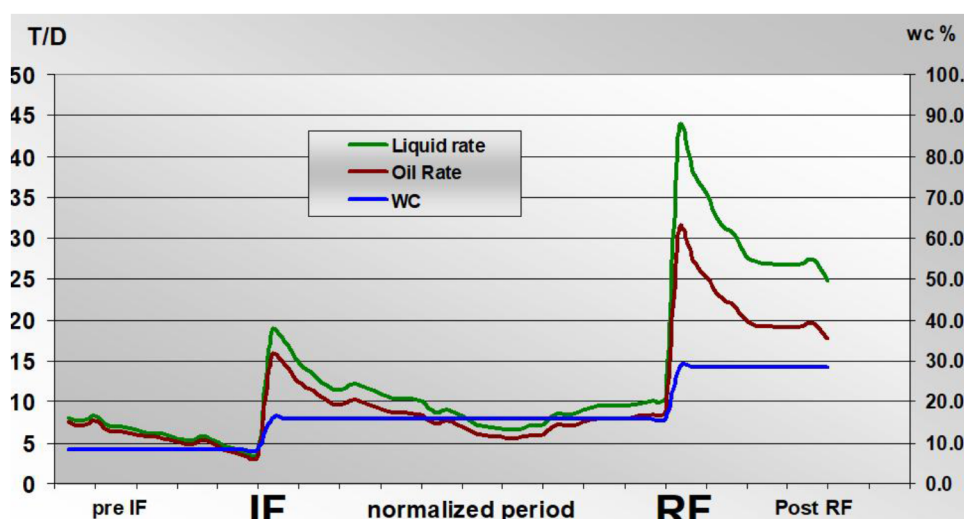


Figure 17—Average production history based on time-normalized scale for 30+ refractured wells. IF- initial fracturing, RF – refracturing.

Conclusions

A comprehensive analysis of the 30+ wells treated with refracturing operations in Jurassic, Achimov and Cretaceous formations in Western Siberia is provided. Study revealed the importance of candidate selection and recognition process to minimize the risk of production underachievement. Main risks encountered for the refracturing operations are:

- Lack of formation pressure because of depletion
- Existing negative skin that is not considered while forecasting the production potential
- Risk of breakthrough into water – either vertical direction or into water injector's front

It is important to use complete production history match as an advantage for refracturing candidates and use representative points in time as equations to resolve uncertain variables such as skin-factor, pressure and kH values for the re-fracturing assessment. Refracturing treatment must be of bigger design volume and must be executed properly to bring substantial productivity increase.

By following these rules and workflows, authors were able to achieve impressive results in majority of the treated wells having less than 1 year payout time of the operations under continuous incremental oil production of 10 tons of oil per day for the duration of several years after the refracturing operations.

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